## GRPO with Unified Rewards

🔥 / ❄ Trainable / Frozen Parameters

$\mathbb{D}_{KL}$: KL Divergence

### Format Reward

$R_{\text{format}} = \textbf{Format}(P_{\text{cot}}, P_{\text{box}})$

Whether the outputs follow the format?

### Box IoU Reward

$R_{\text{box}} = \textbf{IoU}(P_{\text{box}}, GT_{\text{box}})$

How accurate is the box?

### Segment IoU Reward

$R_{\text{seg}} = \textbf{IoU}(P_{\text{seg}}, GT_{\text{seg}})$

How accurate is the segment?

### KL Divergence

$\mathbb{D}_{\text{KL}}(\pi^i_{\text{mllm}}, \pi^i_{\text{ref}})$

How far $\pi^i_{\text{mllm}}$ deviates from $\pi^i_{\text{ref}}$?

## Segmentation

Segmentation Objectiveness & Segment IoU Reward $P_{\text{seg}}$

Segmentation Image Encoder ❄

Decoder 🔥

## Understanding

(Prefilling Tokens)

GRPO Grounding Rewards "I find a cat in arms of a dog" <$x_1$, $y_1$> <$x_2$, $y_2$>

Connector 🔥

MLLM $\pi_{\text{mllm}}$ 🔥

"You are a helpful Assistant. Find the object that best matches…"

" cat"

**Autoregressive Generation** (generated by $\pi_{\text{ref}}$ in training )

**Forward** (for contexts)

System Prompt | Instruction | Image | $P_{\text{cot}}$ | $P_{\text{box}}$ | Context Query 🔥